When Do the Images Prove a Dent?

Identifiability and abstention in CAD-referenced damage estimation for turbine blades

[**Your Name**] Khalifa University October 2026 Target: CVPR 2027 (abstract Nov 10, paper Nov 16, 2026); fallback ICCV 2027

1. Problem

Turbine and compressor blades are inspected from photographs. A dark patch in a photo may be a **dent** (real damage), a **stain**, a **lighting change** or a small **camera error**, and they can look almost identical: a 0.1 mm dent changes a few grey levels, while a 1° camera error changes thousands of pixels. Inspectors need to know whether there is physical damage, where it is, and how large it is, and a method must not invent damage the photos cannot support.

2. Gap in the literature

- **Pose/illumination-agnostic anomaly detection** (PAD, NeurIPS'23; SplatPose, 2024; PIAD, CVPR'25) outputs 2D anomaly maps, not physical 3D change; making a representation illumination-invariant also removes part of the signal a shallow dent produces.
- **CAD-prior reconstruction** (CAD-NeRF, 2024; CADSplat, 2026) fits a deformation to a CAD model to build a digital twin; the deformation absorbs mismatch and is not a tested defect.
- **3D anomaly benchmarks** (MVTec 3D-AD, Real3D-AD, SiM3D) score anomalies without measured depth.

No method outputs a measured 3D change together with a statement of when the photos cannot support it. An author of PIAD, contacted about this work, independently suggested this identifiability-and-abstention question as a meaningful research contribution.

3. Research question

Under explicit acquisition constraints, when is a geometric defect distinguishable from illumination, material or camera variation, and when must a method report that the evidence is insufficient?

4. Approach

Foundation: differentiable mesh rendering (analysis-by-synthesis; nvdiffrast) plus a calibrated statistical test.

- **Input:** healthy CAD mesh + 2–7 calibrated photos. **Output:** *damage supported* with 3D location and depth (mm), or *not supported*.
- **Constraints (assumptions):** calibrated rig (small per-view corrections), fixed light positions with free gains, smooth brightness field, local stain (albedo) patch.
- **Test:** H_0 = nuisance only (pose, lighting, stain) vs. $H_1 = H_0$ + a dent, both fitted through the renderer; dent proposals from a Rao score test; report only if the fit gain exceeds a split-conformal threshold calibrated on healthy and stained blades; otherwise abstain.
- **Detectability:** the same fit predicts, before searching, which dents the views can reveal.

5. Preliminary results (rendered blades only)

Benchmark: 696 cases on public Rotor37 blades + 456 on six industrial blade CAD designs; sealed test sets frozen before evaluation.

Result	Value
Visible dents located at 5% false alarms (30 unseen blades)	**51%** vs. ≤19% for CAD-prior fields (CADSplat-style) and a 2D U-Net
Dents deeper than 0.2 mm found	12 of 13
False alarms on new industrial designs (after fix, v1.1)	51% → **4%**
Textbook tests, actual false alarms at nominal 1%	χ^2 32%, a-contrario 96% (calibration is required)
Predicting which dents are detectable (AUROC)	0.90–0.95 on two sealed tests
Four defect types, held-out industrial blades	dents 23/24, nicks 23/32, cracks 21/32 located; SIFT 16–25%
Depth (measured / true, median)	0.89 Rotor37; **0.60** industrial (systematic bias)

Key finding: presence and location are identifiable and transfer to new blade designs; depth is not. More views raise detection (45 vs. 36 of 60 dents) but not depth accuracy, which points to illumination design rather than viewpoints.

6. Plan: from synthetic to physical blades

1. **Print** industrial blades (from our CAD) with a 3D printer.

2. **Create damage:** dents, nicks, cracks and stains of controlled size; keep healthy and stain-only blades as negatives.
3. **3D scan** every blade: measured ground-truth shape in mm.
4. **Photograph** with a calibrated rig from 2–7 views, under fixed and grazing (low-angle) light.
5. **Compare** the method's output with the scan: detection, false alarms, location and depth error, using the same metrics as on synthetic data.

7. Next experiments

- **Recalibrate** the threshold on industrial blades 1–2 (four-defect false alarms 12.5% → target 5%).
- **Retrain the appearance model** with industrial-blade renders to test the cause of the depth bias.
- **Grazing light:** dents cast shadows, which should make depth identifiable.
- **Multiple views:** oblique views aimed at a suspected region (two-stage capture).
- **Real data:** image alignment (markers/SIFT) and a retry on the SiM3D real benchmark.

8. Timeline

Oct 15–25	Recalibration, appearance-model retraining, grazing-light renders; print and damage blades
Oct 26–Nov 5	Scan and photograph physical blades; run the method; first real results
Nov 6–10	Paper revision (identifiability framing, assumptions, real results); abstract registration Nov 10
Nov 11–16	Final experiments, supplementary material, submission Nov 16 (else ICCV 2027)

9. Expected contributions

1. A formulation of CAD-referenced damage estimation as identifiability with abstention, under explicit constraints.
2. A calibrated render-and-test estimator with a predictor of which damage the views can reveal.
3. Evidence that presence is identifiable across blade designs while depth requires better illumination.
4. A benchmark (synthetic + physical blades with scanned ground truth) for evidence-bounded damage estimation.

10. Resources needed and risks

Needed: 3D printer and material, 3D scanner, calibrated camera rig (2–7 views), controllable lights including one grazing light, GPU time.
Risks: real metal reflections differ from renders (mitigation: calibrate on real healthy blades; matte printed parts first); physical data may not be ready by Nov 16 (mitigation: submit the synthetic + identifiability study, or move to ICCV 2027 with real data).